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Development and evaluation of DASH diet tailored messages for hypertension treatment

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Abstract

For this study, the Manage Associated Perceptions (MAP) of Dietary Behavior Study, researchers developed, evaluated, and tested messages tailored to improve compliance with the Dietary Approaches to Stop Hypertension (DASH) diet by gradually instilling healthy Cognitive Representations of the DASH Diet (CRDDs). The sample consisted of women from diverse backgrounds ($N = 53$), randomly assigned to two experimental groups ($n = 13$ and $n = 14$) and to two control groups ($n = 12$ and $n = 14$). The experimental groups performed a version of the intervention for 30 days. Data about dietary compliance and CRDDs were collected at 30, 60, and 90 days. Compared to control group members, experimental group members demonstrated greater improvements in CRDD scores and were significantly more compliant with the diet. Another significant finding was that older women were more compliant.

1. Introduction

Hypertension (HTN), the "silent killer," is physically devastating because, when uncontrolled, it can cause arteriosclerosis, cerebrovascular accidents, myocardial infarction, and end-stage renal disease (Alderman, 1999). The leading cause of death for women is coronary heart disease, which is often caused or exacerbated by HTN (American Heart Association [AHA], 2004). In the United States today, African Americans, women older than 59 years, and older people have a higher prevalence and, even when treated, a greater severity of HTN than Caucasians, men, and younger people (Hajjar & Kotchen, 2003). National Health and Nutrition Examination Survey (NHANES) data for 1999–2000 shows that 33.5% of non-Hispanic Blacks have HTN and that 71.9% of these fail to achieve optimal therapeutic levels (Hajjar & Kotchen, 2003). It is also well known that there is a direct and positive relationship between age and increased prevalence and severity of HTN. For people aged 60–74 years, 72.6% of the African American population and 50.6% of the Caucasian population had HTN during 1988–1991 (Burt et al., 1995).

Nonpharmacological approaches to HTN treatment are currently being investigated because of the growing recognition that medication only *reduces*, rather than completely obliterates, the risk of future heart problems (MacMahon & Rogers, 1993). For this reason, numerous researchers have turned their attention away from medication-only trials, and instead focused on the more clinically efficacious and cost-effective interventions of dietary change (McCarron, 1998). A sustained reduction of 2 mmHg in diastolic blood pressure (DBP), which is achievable with lifestyle modifications, would result in a 17% decrease in the prevalence of HTN, as well as a 6% reduction in the risk of coronary heart disease, in the elderly (Cook, Cohen, Hebert, Taylor, & Hennekens, 1995). Studies have shown the Dietary Approaches to Stop Hypertension (DASH) diet to be as effective for lowering blood pressure (BP) levels as the daily consumption of one prescription medication (Appel et al., 1997). However,

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long-term adherence to dietary modification is difficult for most people (Little et al., 2004).

There is a need for interventions that help people adhere to dietary modifications. Leventhal, Diefenbach, and Leventhal (1992) linked health behavior to cognitive representations (CRs). CRs are mental images that function to identify, store, and structure environmental stimuli to symbolize beliefs, preferences, and practices (Scisney-Matlock & Watkins, 1999). Accordingly, Scisney-Matlock (1997) modeled how intentions for behaviors that lower HTN are cognitively represented. Her model tracked physiological, psychological, and procedural variables by the use of three cognitive dimensions: knowledge, attitudes, and skills.

A growing number of studies suggest that interventions containing tailored messages can modify existing CRs and thus change health-related behavior (e.g., Campbell et al., 1994). From Rothman and Salovey's (1997) theory of the role of message framing for shaping CRs to improve health-related behavior, it appears that knowledge about specific dietary modifications that is processed through sensory mechanisms can be framed in terms of gains (positive) or losses (negative). Therefore, the first author theorized that positively framed text messages about the DASH diet would create healthy CRs or transform maladaptive CRs, and thus change dietary behavior to conform to the DASH diet. The first step was to establish concrete goals for healthy eating. The DASH diet guided establishment of these goals. The next step was to write messages that led to these goals. The messages were tailored in two ways: (1) to fit any patient's preference for cognitive dimension and (2) to adapt to the culture of the target audience (African Americans, women, and older hypertensives). The goal and message combinations for the skill dimension are listed in Table 1. The cultural sensitivity of the messages is one of the most innovative aspects of this intervention.

The main hypothesis for this study was that this culturally sensitive intervention would produce CRs of DASH dietary behavior (CRDDs) that would gradually improve adherence to the DASH diet, especially in African American women. Additional hypotheses were that women in the experimental group would exhibit higher CRDD scores than women in the control group and that older women would exhibit higher CRDD scores and better dietary compliance than younger women.

Table 1
Phase 1: Goals (in bold) and skill dimension messages

1. **Eat 4–5 servings of fruit a day.** Try fresh fruits in season/make a fruit salad of "light" or "natural juice" canned fruits. Keep a mixture of dried fruits in your desk or car for "emergency" snack attacks.
2. **Eat 7–8 servings of grain a day.** Check out and try new/unusual grains like Japanese soba noodles or kasha, in addition to common staples like pasta, rice, cereals, and bread.
3. **Eat 4–5 servings of vegetables a day.** Place your vegetables and grains in the center of your plate and place meat on the side or on a smaller plate, e.g., your low-fat beef burger, meatloaf, chicken, fish, etc.
4. **Eat 2–3 servings of low-fat or nonfat (skim) dairy foods.** Use "light" yogurt (or Lactaid-treated skim or 1/2% milk if you have trouble digesting milk) to make puddings, soups, custards, gravies, and sauces.
5. **Eat up to 2 servings of lean meat, fish, or poultry per day.** Choose portions of meat about the size of a deck of cards. If a portion is larger, cut off that which exceeds what you need for the DASH diet. Save it for another meal.
6. **Eat 4–5 servings of nuts, seeds, or dried beans per week.** Eat meatless meals containing peas, beans, or lentils several times a week. Try vegetarian chili, soup, or fat-free "refried" beans in burritos.
7. **Limit use of salt from the shaker to no more than 2,400 mg a day.** Read labels carefully! Ask your dietitian to explain how to understand them. Make sure what the label calls a serving is the same amount you eat. Replace salt with curry, spices. herbs.
8. **Use/eat fats sparingly.** Select meats, fish, poultry with no more than 3 g of fat per ounce (10% fat by weight). Butter, cheese, and ice cream are high in saturated fat.
9. **Use/eat sweets in moderation.** Savor and enjoy every bit of your food! Stop when you are satisfied. You can always have more later. Aim for no more than 5 servings a week of sweets, including chips, pie.
10. **Choose healthy snacks, such as fresh fruits and raw vegetables.** Eat fruit, yogurt, fat-free pudding, small amounts of nuts, crisp vegetables with fat-free dip, unsalted pretzels, reduced fat crackers (Graham crackers) or unsalted popcorn.
11. **Eat at least 25 g of fiber in food (whole grains) for "roughage" per day.** Figure out ways to increase fiber in your diet. Use whole-grain bread, pasta, and brown rice. Add lots of fruits, dried beans, and vegetables to increase your fiber.
12. **Sit down for each meal including breakfast!** Skip your evening meals for a few nights and you might wake up hungry! Eating late does not give you time to feel hungry by breakfast.
13. **Do some physical activity every day that requires you to think about using up calories.** Try to exercise 20 minutes a day at least 3 days a week. Try walking, taking the stairs, jogging, bicycling, etc. For a gift, request a set of "light" hand weights.
14. **If you drink alcohol, drink in limited amounts and don't drink to please others.** Don't add extra calories to your diet with alcohol. One drink is equal to 1 1/2 oz of 80 proof whiskey, 5 oz of wine, or 12 oz of beer.
15. **Maintain a healthy weight for your height and level of activity.** Focus on lifestyle changes for long-term healthy eating, not dieting. Following the DASH diet will improve your health. Set your eating goals. Do one thing each day to make it more successful than yesterday.
16. **Consult your dietitian, doctor, and/or nurse (any health care provider) if you have questions.** Make a list of questions/concerns you have about the DASH diet. Have questions ready every time you meet with your health care provider.
17. **Follow the DASH diet!** Changing how you eat is very difficult, even when you know what you need to do. Break down a difficult problem into small steps so it's easy to overcome. Recognize and celebrate your success. Write down what was hard and easy to change and why.
18. **Use a good low-fat cookbook to help you to follow a healthy diet!** Change your eating habits to lower blood pressure. Eating is under your control and is the best way known by man to prevent and control it.

This article documents the development, evaluation, modification, and testing of the tailored message intervention and the relationships between selected demographic variables and CRDDs and compliance.

2. Methods

During this study, the Manage Associated Perceptions (MAP) of Dietary Behavior Study, researchers developed and tested tailored messages to increase compliance to the DASH diet. The study was approved by all applicable committees to safeguard human subjects. All participants signed informed consent forms.

There were five phases to this study. They are described in the Procedures subsection.

2.1. Study design

This research included a randomized, single-blinded, and patient-centered study that used the intent-to-treat design. Study group assignments were random, and study participants were unaware of their group assignments. Because researchers provided one of the experimental groups, but not either control group, with bar charts of their baseline CRDDs, the study was not blinded to the researchers. Data analysis included comparisons between these study groups and between additional groups based on demographic characteristics as well as within-group comparisons over time.

To determine whether there were any ambiguities or difficulties that participants might encounter during the trial, researchers selected five participants as a convenience sample for a pilot study to pretest the intervention. After pretest measures, the five participants were exposed to the study protocol. Participants in this pilot study performed all of the same activities, including taking the CRDD survey, requested of experimental group participants in the actual study.

2.2. Sample description

Fifty-three women participated in the study. All had systolic blood pressure (SBP) ≥ 140 mmHg and/or DBP ≥ 90 mmHg or were taking anti-HTN medicine. Only women were studied because men and women differ in the prevalence, severity, control rates, and age of onset of HTN (Burt et al., 1995). Because African Americans surpass Caucasians in the prevalence and severity of HTN and lag behind Caucasians in control rates (Hajjar & Kotchen, 2003),

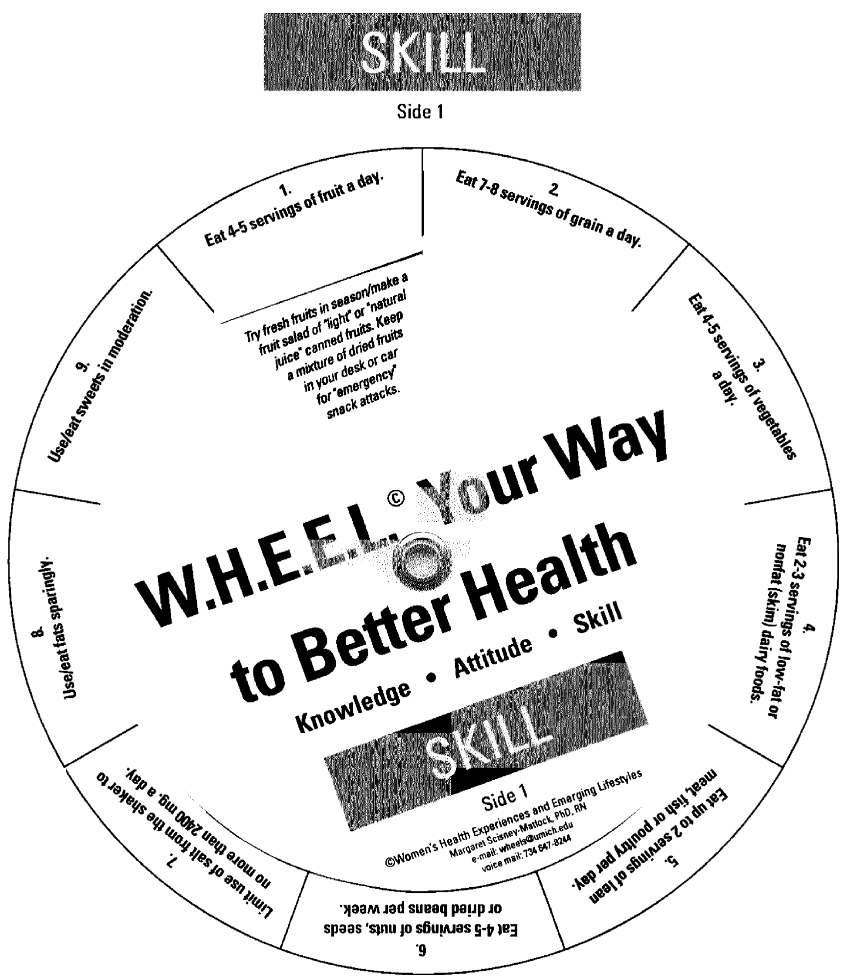

Fig. 1. The Manage Associated Perceptions (MAP) wheel for the skill dimension (Phase 4).

researchers recruited more than 40% minorities. The minority ethnicity category contained 91% African American women.

After eligible participants read and signed the consent form and provided baseline demographic data, researchers selected participants. Then, researchers used a computer program to randomize participants to either one of two experimental groups or one of two control groups and scheduled them for orientation. Group assignment was determined in numerical sequence from 48 numbered envelopes matching the stratified sampling criteria representative of a Solomon Four-Group Design. Thus, the randomization procedures were designed to ensure a racially diverse sample that met the requirements for a Solomon Four-Group Design. Each group had a minimum of 12 participants. To maintain the minimum of 48 participants, researchers replaced any women who withdrew after orientation and assignment.

Ultimately, 53 hypertensive women provided data for analysis. Most were more than 63 years of age and married; 50% had a college education. Researchers classified the entire sample according to two characteristics to determine the influence of each characteristic on the effectiveness of the intervention: age (46–62 and 63–80 years) and ethnicity (minority and Caucasians).

2.3. Intervention

The two experimental groups employed either the DASH diet only or the DASH diet and exposure to a home-based program of reading DASH diet tailored messages; both provided monthly data. The two control groups provided data either at the end of the study only or monthly and at the beginning and end of the study. Experimental group participants performed an intervention for 30 days. The intervention consisted of (1) an introduction to the DASH diet, and possibly, (2) the physical devices that held the MAP messages, and (3) the actions and tasks patients performed with the intervention devices. To create the physical devices, the first author made three separate paper wheels, one for the knowledge dimension, one for the attitude dimension, and one for the skill dimension (Fig. 1). All three wheels contained the same set of 18 goals around the circumference, 9 on one side and 9 on the other. For each goal, each wheel contained one message framed to support that goal in the information dimension of that wheel. On each side of each wheel, there was a round piece of cardboard attached with a window cut out. When a patient turned the cardboard circle until the window was below a goal, the message for that goal, in that dimension, appeared in the window.

Experimental group participants were given notebooks for recording thoughts and feelings and daily food intake. Participants in the first experimental group (those required to perform the extended intervention) were also given the three wheels and a bar chart displaying their baseline CRDD scores. Each was instructed to do the following every morning for a 30-day period: review an unhealthy CRDD from her bar chart, use the wheels to view the three messages for that goal, and record in the notebook the feelings and thoughts she has about the goal she selected. For example, if the goal chosen by the participant was "Eat 4–5 servings of fruit a day," she might write, "I am learning to appreciate the sweetness of raw fruit." In the evening, she

Table 2
Phase 2: Average scores among raters for "representativeness" (Rep) and "ease of interpretation" (Ease) of messages by dimension

Corresponding DASH diet goal	Dimension					
	Knowledge		Attitudes		Skills	
	Rep	Ease	Rep	Ease	Rep	Ease
1. Eat 4–5 servings of fruit a day.	1.6	2.2	3.0[a]	3.2[a]	1.4	1.6
2. Eat 7–8 servings of grain a day.	2.2	2.4	1.8	1.6	2.8[a]	2.2
3. Eat 4–5 servings of vegetables a day.	1.4	2.0	1.8	1.6	2.2	1.4
4. Eat 2–3 servings of low-fat or nonfat dairy foods.	1.8	1.8	2.6[a]	2.2	1.8	2.0
5. Eat up to 2 servings of lean meat, fish, or poultry per day.	1.8	2.6[a]	3.0[a]	2.8[a]	1.7	1.8
6. Eat 4–5 servings of nuts, seeds, or dried beans per week.	1.6	2.4	2.0	2.5[a]	1.6	1.8
7. Limit use of salt from the shaker to no more than 2,400 mg a day.	2.0	3.0[a]	2.2	2.8[a]	1.7	1.6
8. Use/eat fats sparingly.	2.2	3.0[a]	2.6[a]	2.8[a]	2.2	2.4
9. Use/eat sweets in moderation.	2.4	2.8[a]	2.4	2.0	2.2	2.0
10. Choose healthy food snacks, such as fresh fruits and raw vegetables.	1.8	2.2	1.6	1.8	1.6	1.6
11. Eat at least 25 g of fiber in food (whole grains) for "roughage" per day.	2.0	2.6[a]	1.8	2.2	2.4	1.6
12. Sit down for each meal including breakfast!	1.8	1.8	2.4	2.0	3.2[b]	1.8
13. Do some physical activity every day that requires you to think about using up calories.	2.6[b]	2.4	2.2	2.4	2.2	1.4
14. If you drink alcohol, drink in limited amounts and don't drink to please others.	1.7	2.0	3.0[b]	2.0	2.4	1.8
15. Maintain a healthy weight for your height and level of activity.	2.1	2.5[b]	2.0	1.6	1.6	2.0
16. Consult your dietitian, doctor and/or nurse (any health care provider) if you have questions.	2.0	2.2	2.0	2.2	2.0	1.4
17. Follow the DASH diet! Changing how you eat is very difficult, even when you know what you need to do.	1.6	2.0	1.4	1.4	1.6	1.6
18. Use a good low-fat cookbook to help you follow a healthy diet!	2.0	1.8	2.6[b]	1.4	1.8	1.8

[a] Mean ≥ 2.5 = an invalid item.
[b] Mean ≥ 2.5 = an invalid item.

recorded performance of the intervention activities, along with everything she had eaten that day, in the notebook. The extended intervention took the longest, about 20 minutes.

Participants in Experimental Group 2 followed the DASH diet and recorded their activities and diet in the notebook. They did not review charts of their baseline CRDDs or use the wheels. The control groups did nothing but take the CRDD survey form and the Health Promotion Lifestyle Profile (HPLP; one control group at 30, 60, and 90 days and the other control group only at 90 days).

2.4. Instruments

2.4.1. Survey Form for the Evaluation of DASH Diet Messages

As part of Phase 2, the first author developed the Survey Form for the Evaluation of DASH Diet Messages for health care professionals to use to assess the messages. The form contained two questions about each message: "Does the information in the message represent the DASH diet?" and "Is the message easy to interpret and understand?" There were four possible responses to each question, with 1 and 2 being the positive scores. Responses numbered 3 or 4 indicated that the message was lacking in either representativeness or ease of interpretation or both. A mean score was calculated for each aspect of each message by taking the average of the ratings by the health care professionals. Scores for all three dimensions could range from 108 to 432, with lower scores indicating higher assessments of the messages. See Table 2 for the mean scores for the 18 goals by information dimension. From these ratings by health care professionals, a content validity index (CVI) was calculated for each aspect (i.e., representativeness and ease of interpretation) of each message. To do this, researchers divided the number of messages considered valid ($\leq$2.5) for each aspect by 54 and multiplied the result by 100. Messages were then rewritten as necessary.

2.4.2. The Health Promotion Lifestyle Profile

To determine each participant's actual dietary behavior, researchers used the HPLP survey (Walker, Sechrist, & Pender, 1987). The survey was designed to determine the frequency of specific health behaviors. Compliance to the DASH diet, the dependent variable, was quantified from scores on the nutrition behavior section, which consisted of 30 behaviors. Each person used a 4-point scale to estimate how often she engaged in the behavior. Scoring was 1 = *never*, 2 = *sometimes*, 3 = *often*, and 4 = *routinely* for positive behaviors and the reverse for negative behaviors (Walker et al., 1987). Scores on the Nutrition subscale could range from 30 to 120, with higher scores indicating better dietary behaviors. The HPLP survey has been shown to be reliable; the alpha coefficient for the Nutrition subscale is .76 (Walker et al., 1987).

2.4.3. The CRDD scales

Using the Lifestyle Cognitive Representations Scales (Scisney-Matlock, 1998) as an example, the first author developed the CRDD scales to assess CRDDs. For each DASH diet goal, the participant selected one of six responses to each of three assertions or questions: "Describes me now.," "How much does it matter?," and "Likely to describe me in the future." (See Table 3 for an example question.) The selected responses to these three assertions or questions represented the three cognitive dimensions (knowledge, attitude, and skills) of the goal. Separately, higher scores represented more accurate CRs for that dimension. Higher mean scores for the three dimensions calculated together indicated a more accurate CR for that goal. Overall scores for the CRDD scales could range from 0 to 270, with 270 representing entirely accurate CRDDs.

2.5. Procedures

This study consisted of the following five phases:

1. Development of the goals and messages to instill healthy CRDDs.
2. Assessment by health care professionals to determine whether the messages could form the desired CRDDs in the target population.
3. Modification of the messages based on the professionals' assessments and creation of the CRDD scales.
4. Development of an intervention that incorporated the modified goals and messages.
5. Testing of the intervention with a sample containing a diverse group of hypertensive women.

2.5.1. Phase 1: Development of the necessary goals and messages

Working together, the first and second authors examined the official dietary recommendations for treating HTN with the DASH diet. They then discussed the issues and challenges in treating HTN with diet with a convenience sample of hypertensive patients from various ethnicities. From this information, they determined what CRDDs would lead to a healthy diet for hypertensives. They then developed 18 goals that would form those healthy CRDDs.

Table 3
Phase 3: Sample item for the CRDD Scale

1. Eat 4–5 servings of fruit a day.						
a. Describes me now.	Not at all	A little	Somewhat	Quite a bit	Very much	Don't know
b. How much does it matter?	Not at all	A little	Somewhat	Quite a bit	Very much	Don't know
c. Likely to describe me in the future.	Not at all	A little	Somewhat	Quite a bit	Very much	Don't know

Table 4
Phase 5: Stratification of sample and Solomon Four-Group research design (N = 48 or more)

Solomon Four-Group	Treatment assignment	Minority (n)	Caucasian (n)
Experimental Group 1	DASH diet intervention, pre- and posttest	6 or more	6 or more
Experimental Group 2	Extended intervention, pre- and posttest	6 or more	6 or more
Control Group 1	Pre- and posttest	6 or more	6 or more
Control Group 2	Posttest	6 or more	6 or more
Subtotals		24 or more	24 or more

For each goal, they wrote three messages, one for each dimension, using the Fog test to ensure that the text was at an eighth-grade reading level.

2.5.2. Phase 2: Assessment by health care professionals

Five health care professionals were asked to rate each message on representativeness and ease of interpretation. These professionals, four registered dietitians and one clinical nurse practitioner, were a convenience sample. All had expertise in clinical nutrition and previous experience with teaching hypertensive patients about the benefits of dietary modification. The health care professionals were given the complete DASH diet, the 54 MAP messages, and the Survey Form for the Evaluation of DASH Diet Messages.

2.5.3. Phase 3: Modification of the messages

Researchers analyzed the above assessments to determine which messages needed to be rewritten and how they needed to be rewritten. First, they calculated the mean score for each dimension of each message. It was decided that mean scores ≥ 2.5 would invalidate a message. Of the 108 total items (i.e., 18 messages rated along the three dimensions of knowledge, attitudes, and skills for both representativeness and ease of interpretation), 88 were rated as valid. After analyzing the raters' responses, the first author rewrote the messages according to the assessments and created the 18 items on the CRDD scales from the goals.

2.5.4. Phase 4: Development of an intervention using the modified messages

Researchers created an intervention that would enable people to focus on the 18 goals and the associated messages.

2.5.5. Phase 5: Testing the intervention on a diverse group of hypertensive women

Researchers devised a study based on a Solomon Four-Group design (Table 4) to test the intervention on a diverse sample of hypertensive women. Researchers recruited participants from an outpatient clinic at a large Midwestern university hospital, from advertisements placed in university employee publications, or through personal contacts by study participants who recommended that other women with HTN join the study (Table 5). After selection and baseline data collection, participants were randomly assigned to experimental or control groups. Researchers then tested the intervention's effect on CRDDs and compliance with the DASH diet. Outcome measures were the CRDD scores that assessed CRDDs and the HPLP (Walker et al., 1987) scores that assessed dietary behaviors or compliance. Outcome measures were assessed at 30, 60, and 90 days after the start of the intervention.

2.6. Data analysis

Using compliance to the DASH diet (i.e., HPLP scores) as the dependent variable and the CRDD scores as dependent testing variables and sociodemographic factors of interest as independent grouping variables, researchers performed t tests on independent samples. In addition, statistical tests were performed with age as a covariate in repeated measures ANOVA (ANCOVA) to ascertain whether it influenced compliance or CRDD scores over time. Tests were performed for each dimension across time for the entire CRDD scale, for the HPLP survey, and for a combination of both; these tests established internal consistency. Lastly, correlational analyses determined which of the three dimensions of CRDDs was most strongly associated with enhanced compliance.

3. Results

During Phase 3, researchers found that the CVI results, which quantify validity, were highest for ease of interpretation for messages in the skill dimension (100%). Following Lynn (1986), a CVI of 82% or more was considered acceptable. Representativeness for messages in the knowledge dimension (94.4%) and representativeness for messages in the skill dimension (88.9%) also had acceptable CVIs. Messages in the attitude dimension had low CVIs for both representativeness (66.7%) and ease of interpretation (72.2%). The overall CVI was 90% for representativeness and 80% for ease of interpretation. The overall CVI for both aspects of both dimensions was 83.3%.

During Phase 5, researchers collected and analyzed data from a sample of hypertensive women. Because responses to the CRDD scale were similar to those for the nutritional part

Table 5
Phase 5: Sample recruitment

Source of recruits	Total	Ineligible	Eligible	Declined	Withdrew	Participated
The University of Michigan Hypertension Clinic	76	4	72	19	8	45
Advertisements	10	2	8	5	1	2
Personal contacts	6	–	6	–	–	6
Total	92	6	86	24	9	53

Table 6
Phase 5: Baseline demographics for Solomon Four-Group assignment

Characteristic	Experimental Group 1 (n = 13), n (%)	Experimental Group 2 (n = 14), n (%)	Control Group 1 (n = 12), n (%)	Control Group 2 (n = 14), n (%)	Total sample (N = 53), n (%)
Age (years)					
46–62	6 (11.3)	8 (15.1)	7 (13.2)	5 (9.4)	26 (49.0)
63–80	7 (13.2)	6 (11.3)	5 (9.4)	9 (17.0)	27 (50.9)
Ethnicity					
Caucasian	7 (13.2)	7 (13.2)	8 (15.1)	8 (15.1)	30 (56.6)
Minority	6 (11.3)	7 (13.2)	4 (7.5)	6 (11.3)	23 (43.4)
Education					
>High School	2 (3.8)	1 (1.9)	3 (5.7)	2 (3.8)	8 (15.1)
Some college	5 (9.4)	3 (5.7)	3 (5.7)	7 (13.2)	18 (34.0)
College degree	6 (11.3)	10 (18.9)	6 (11.3)	5 (9.4)	27 (50.9)
Marital status					
Married	9 (17.0)	5 (9.4)	10 (18.9)	8 (15.1)	32 (60.4)
Not married	4 (7.5)	9 (17.0)	2 (3.8)	6 (11.3)	21 (39.6)
Study group					
Experimental	13 (24.5)	14 (26.4)	–	–	27 (50.9)
Control	–	–	12 (22.6)	14 (26.4)	26 (49.0)

of the HPLP survey, researchers felt safe in assuming that the two measures tapped into the same CRs of lifestyle change behaviors. Internal consistency tests revealed Cronbach's alpha (α) of .89 (std item α = .888) for the CRDD scale, α .91 (.919) for the HPLP survey, and α .82 for the combined CRDD scales and HPLP survey. Internal consistencies of the three dimensions across time were .85 (.923) for knowledge, .86 (.938) for attitude, and .85 (.929) for skill.

To determine the effects of age and ethnicity on CRDD and HPLP scores, researchers first examined baseline data. (See Table 6 for baseline demographic information.) Researchers obtained the baseline demographic characteristics of the sample from questionnaires and BP measurements. Participants ranged in age from 46 to 80 years. The sample was well educated; over 50% had a college degree or higher. A majority (31 out of 53) had their BP under control at baseline. There were no significant differences between the experimental and control groups in either CRDD scores or compliance (i.e., HPLP survey scores) at baseline. When the three dimensions (knowledge, attitude, and skill) of the CRDDs were taken as an aggregate, there were no significant differences between participants by any demographic variable. For the individual dimensions of knowledge and skill, there were no significant differences by any demographic variable. However, at baseline the minority group had significantly higher scores than Caucasian women on attitude toward DASH dietary practices (M = 22.17 and 25.64, respectively; t = −2.161; p = .037).

Significant main effects in mean CRDD scores were observed between groups across time (Table 7). On the

Table 7
Phase 5: Influence of selected characteristics on CRDD scores over time

Group	Time (days)	M	SD	MSE	t	df	p
Study group							
Experimental	30	76.70	10.77	1.10	1.84	16.1	<.08
	60	76.15	10.20	1.96	3.12	39.0	<.003[a]
	90	75.80	8.81	1.67	3.64	33.9	<.001[a]
Control	30	56.21	21.92	5.86	1.84	16.1	<.08
	60	63.79	15.05	4.02	3.12	39.0	<.003[a]
	90	61.32	18.07	3.61	3.64	33.9	<.001[a]
Ethnicity							
Caucasian	30	60.52	18.82	3.76	−1.67	41.0	<.01[a]
	60	71.75	15.45	3.15	−0.10	39.0	<.92
	90	68.97	17.08	3.02	−0.06	53.0	<.95
Minority	30	68.58	9.49	2.24	−1.67	41.0	<.01[a]
	60	72.18	9.90	2.40	−0.10	39.0	<.92
	90	69.24	13.11	2.73	−0.06	53.0	<.95
Age (years)							
<60	30	62.23	13.30	2.97	−0.64	41.0	<.53
	60	68.11	14.70	3.46	−1.66	39.0	<.10
	90	69.53	17.11	3.83	0.16	53.0	<.87
≥ 60	30	65.35	18.15	3.79	−0.64	41.0	<.53
	60	74.91	11.52	2.40	−1.66	39.0	<.10
	90	68.83	14.61	2.47	0.16	53.0	<.87

[a] Statistically significant.

Table 8
Phase 5: Improvements over time in CRDD scores by dimension and study group

Dimension	60 Days						90 Days					
	M	*SD*	*MSE*	*t*	*df*	*p*	*M*	*SD*	*MSE*	*t*	*df*	*p*
Knowledge				2.53	17.6	<.021[a]				3.74	39.3	<.001[a]
Experimental	22.15	4.45	0.86				22.54	3.66	0.69			
Control	16.50	7.69	2.06				17.44	5.87	1.17			
Attitude				2.16	39.0	<.037[a]				2.78	35.4	<.009[a]
Experimental	27.15	4.31	0.83				26.75	3.96	0.75			
Control	23.57	6.22	1.66				22.08	7.54	1.51			
Skill				2.62	39.0	<.012[a]				3.17	34.7	<.003[a]
Experimental	26.85	3.43	0.66				26.75	3.40	0.64			
Control	23.57	6.22	1.66				21.80	6.70	1.34			

[a] Statistically significant.

whole, CRDD scores changed significantly as a function of study group at 60 and 90 days. Researchers also found substantial growth for both older and younger women when examining CRDD scores as a whole. The older women surpassed the younger women at 60 days (by 6.80 points), but the younger women slightly surpassed their counterparts at 90 days.

Improvements on the separate dimensions were indicated by statistically significant changes in mean scores (Table 8). At 60 and 90 days, means for all three cognitive dimensions were higher for the experimental group than for the control group. For the knowledge dimension, young women's scores grew from $M = 15.58$, $MSE = 1.07$, at 60 days to $M = 20.25$, $MSE = 1.32$, at 90 days, whereas older women's scores changed from $M = 18.33$, $MSE = 1.21$, at 60 days to $M = 20.34$, $MSE = 0.88$, at 90 days. By 60 days, a main effect occurred between the two age groups for the knowledge dimension when $t = -1.97$, $df = 39$, and $p = .05$. Although young and old women shared similar attitudes at baseline and both groups showed improvement by 60 days, the younger group was more successful at maintaining this change by the end of the study (from $M = 25.22$, $MSE = 1.22$, at 60 days to $M = 25.20$, $MSE = 1.12$, at 90 days for the younger women vs. $M = 26.48$, $MSE = 1.22$, at 60 days to $M = 24.17$, $MSE = 1.05$, at 90 days for the older women). For the skills dimension, younger women's CRDD scores were nearly identical to the older women's by 90 days ($M = 24.08$, $MSE = 1.39$, for the younger women vs. $M = 24.31$, MSE 0.90, for the older women).

There were no clinically or statistically significant effects of ethnicity on CRDD scores over the course of the study. Caucasian women initially had lower mean CRDD scores than African American women, but caught up by 90 days ($M = 24.69$ and $M = 24.35$, respectively).

There were clinically significant, but not statistically significant, differences in compliance (Table 9) according to study group at 60 days ($M = 14.04$, $SD = 3.05$, $MSE = 0.61$, for the experimental group vs. $M = 12.18$, $SD = 3.49$, $MSE = 0.74$; $t = 1.95[45]$; $p = .057$ for the control group). Age, too, affected compliance. Older women consistently demonstrated more compliance as compared with younger women, but the difference was significant only at 90 days. As with CRDD scores, compliance scores were not significantly affected by ethnicity over the course of the study.

At all three periods, the knowledge dimension was strongly correlated with compliant behaviors ($r = .66$, 0.57, 0.70, with all three statistically significant). Attitude correlated with levels of compliance ($r = .38$, statistically significant) at 30 days, was not influential at 60 days, but again correlated with levels of compliance at 90 days ($r = .46$,

Table 9
Phase 5: Influence of study group and age on HPLP (compliance) scores over time

Group	Time (days)	*M*	*SD*	*MSE*	*t*	*df*	*p*
Study group							
Experimental	30	12.07	4.08	0.78	0.18	37.0	<.086
	60	14.04	3.05	0.61	1.95	45.0	<.057
	90	13.96	2.57	0.49	1.58	39.0	<.123
Control	30	11.83	3.49	1.01	0.18	37.0	<.086
	60	12.18	3.49	0.74	1.95	45.0	<.057
	90	12.38	3.75	1.04	1.58	39.0	<.123
Age (years)							
<60	30	10.47	3.50	0.85	2.29	37.0	<.028[a]
	60	11.80	3.71	0.96	1.81	47.0	<.077
	90	12.16	3.08	0.71	2.76	39.0	<.009[a]
>60	30	13.18	3.78	0.80	2.29	37.0	<.028[a]
	60	13.65	3.10	0.53	1.81	47.0	<.077
	90	14.59	2.58	0.55	2.76	39.0	<.009[a]

[a] Statistically significant.

statistically significant). Skills were not significantly correlated with compliance.

4. Discussion

The results of this study indicate that the MAP intervention helped to correct inaccurate CRDDs and to improve compliance with the DASH diet in all groups. The fact that there were no significant differences at baseline between the experimental and control groups in either CRDD or HPLP scores strongly indicated that study group assignment was indeed random, and attested that any change across time could be specifically attributed to the intervention as opposed to preexisting differences. Because there were significant correlations between CRDD scores and compliance scores at every point in time, it appears that the tailored messages promoted long-term adherence to the DASH diet.

The finding that CRDDs and eating habits initially mattered more to African American women was not at all surprising given that Kumanyika (1997) contended that reductions in risk factors due to lifestyle modifications are potentially greater for African Americans than for Caucasians. Although Caucasian women may not have found the cost/benefit ratio worthwhile, perhaps African American women perceived that the benefits were much greater for them, whereas the cost stayed the same.

The hypothesis that older women would exhibit higher CRDD scores than younger women was confirmed for 30 and 60 days by both the sum total of scores for the three dimensions and the individual scores. Older women had higher CRDD scores at 30 days than younger women probably because they had more skills (i.e., "tools") to manage HTN. However, at about 60 days, older women began to lose some of the new tools they had added. These women might have found it difficult to add a tool without first identifying why it was useful. Fortunately, they had already discovered what skills they could implement to motivate themselves to engage in compliant action. Conversely, younger women, who did not have as extensive a "toolbox," might have processed the new information presented to them by identifying and internalizing it so they could file it away for future reference. This task would have preceded compliance and it may have taken longer than the length of the study. That may explain why younger women matched, and even slightly exceeded, older women in CRDD scores by the end of the study, but never matched older women in compliance levels.

One limitation of this study was that it did not address the possibility that some patients had an easier time implementing lifestyle changes in their daily routines because of familiarity with measures to be taken to care for long-term illnesses or problems. Additional studies could include recruitment criteria assuring that all participants have similar experience in using dietary modifications to improve their health.

There need to be more studies of the differences between the rates of learning and learning capacities of the novice compared to the experienced hypertensive patient. There also needs to be further research into the use of tailored messages to change health behaviors, such as medication compliance, adherence to exercise routines, and smoking behavior.

The results of this study support the value of cognitive theory for research into illnesses and various types of health maintenance behavior. Additional research on interventions designed to address initial differences in CRs due to ethnicity and age may enable researchers to better tailor messages to patients' psychological and physical needs. Of course, longitudinal research is needed to elucidate the nature of CRDDs over longer periods. A major value of this study is that it provided the foundation to address dietary modifications for early treatment for HTN.

In this study, all women receiving the intervention displayed improvements in levels of knowledge, attitudes toward disease, and skills to be more compliant. If tailored messages similar to these created for the DASH diet could be developed for other disorders and diseases, patients could quickly learn about their illness and the necessary behavioral changes once they were given some basic instructions. This study shows that short, simple, clear messages such as those used as part of this intervention are effective means of accomplishing both cognitive and behavioral changes.

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